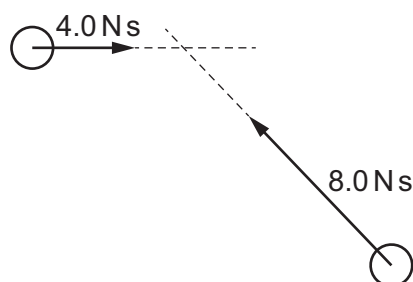
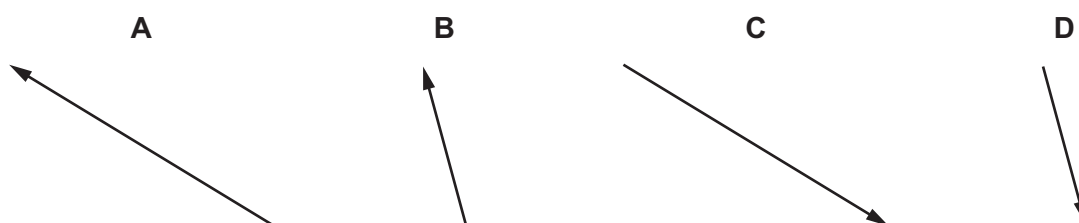


- 10 The diagram shows the view from above of two balls moving along a horizontal frictionless surface before they collide. The momentum of each ball is also shown.

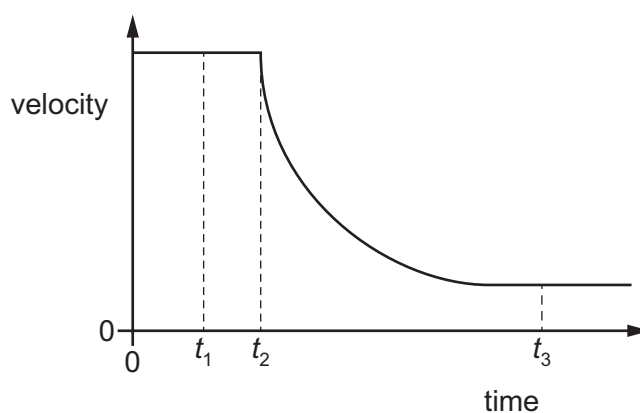


The balls stick together during the collision.

Which vector diagram represents the combined momentum of the balls after the collision?



- 11 A skydiver is falling at constant velocity. She then opens her parachute. The graph shows the variation with time of her velocity.



Which statements about the motion of the skydiver are correct?

- 1 The magnitude of the acceleration is maximum at time t_2 .
- 2 The magnitude of the drag force at time t_1 equals the magnitude of the drag force at time t_3 .
- 3 The magnitude of the drag force is maximum at time t_1 .

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only